Chapter 3

Learning and Evaluation for Mechanistic Modeling

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We have seen in the earlier chapter how to use stochastic modeling (the Erlang M/M/c/k systems) and the resulting Erlang formulas to guide decisions around how many processing resources to allocate for a given level of throughput and probability of delay. The discussion there, however, was predicated on accepting that the model was already determined and correctly captured reality. In this chapter, we will look at how to learn a mechanistic congestion model from data, and the questions that arise concerning the eventual effectiveness of this learned model in guiding actions.

The types of congestion models proposed by Erlang will remain the main character in our story. As a matter of convenience, we will refer to this class of mechanistic stochastic models, first popularized by Erlang, as "Erlangian" models.

3.1 Basics of Learning the Congestion Model

In the case of Erlangian congestion modeling, we implicitly, via human expertise, frame the problem by adopting a mechanistic stochastic model that depicts the congestion dynamics arising from serving randomly arriving requests. Each individual model within this family is further specified by several parameters. By learning a congestion model, we mean identifying the most appropriate combination of these parameters. They include:

1. The arrival rate λ: how many jobs arrive in the system per unit of time?
2. The service rate μ: how many jobs can each server or worker process per unit of time?
3. The number of servers or workers c.
4. The total number of jobs the system can contain, k. Alternatively, the size of the waiting buffer, $k - c$.

Out of these parameters, c and k are usually not considered something to be learned because they can be chosen by the designer themselves. This leaves determining the values of λ and μ to be learned from data.

Cold- vs. Warm-starts

In the case of a call center or data center, λ and μ are typically estimated using operational data from existing operations. The emphasis here is that we are assuming that *some* form of the system the model aims to capture has already been deployed in practice for long enough to collect such data. We call this the *warm-start* modeling problem, where the system is already in operation, affording modelers like Erlang some readily available data to get started. In contrast, there are equally many applications, if not more prevalent, in which the decision-maker faces a *cold-start* problem, where we hope to use modeling to guide the design of a system that has yet to be launched or has only recently been launched.

The detailed statistical procedures employed to estimate and learn these parameters from data can vary greatly depending on the preference of the designer (Bayesian, Frequentist, etc.). As a baseline for our example, things are really quite simple. By choosing such a relatively simple queueing model, we are effectively baking in the assumption that the arrival and service rates are time-invariant, and in particular, that they do not depend on the day of week or time of day. With this restriction in place, suppose that we are learning parameters for a call-center model; estimating λ simply amounts to:

1. Define a sufficiently long data collection horizon, T. Say we let this be one year, so $T = 525,600$ minutes.

2. Measure the number of incoming calls during this year, N. Suppose this number came to be $N = 28,598,331$ calls.

3. Then λ is estimated to be $N/T = 54.41$ calls per minute.

The estimation of μ can be accomplished in a similar manner, by taking the average of processing times of jobs over a sufficiently long time horizon.

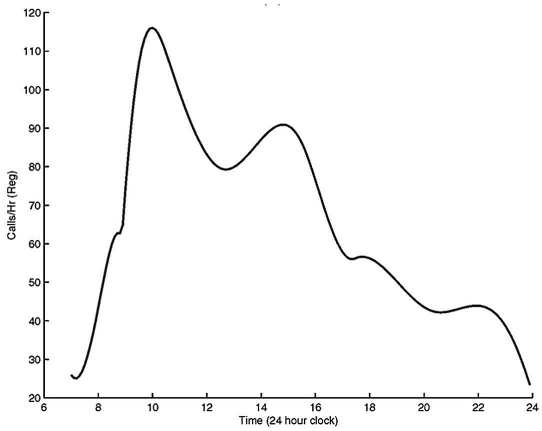
To be more precise, the estimation procedure above corresponds to the frequentist maximum-likelihood estimator for λ under the time-invariance assumption. Some practitioners may also prefer a Bayesian procedure, where they can incorporate stronger prior beliefs about what λ should be. In this example, a large number of observations (across a whole year) is collected to estimate a relatively large arrival rate, and as such these two approaches are unlikely to yield different conclusions. But in cases where arrivals occur sparingly, such as when estimating dynamics related to highly specialized equipment in the METRIC model discussed in Chapter 2, the practitioner's prior could play a much more significant role.

3.2 Evaluating the Erlang Model: A Study with Call Centers

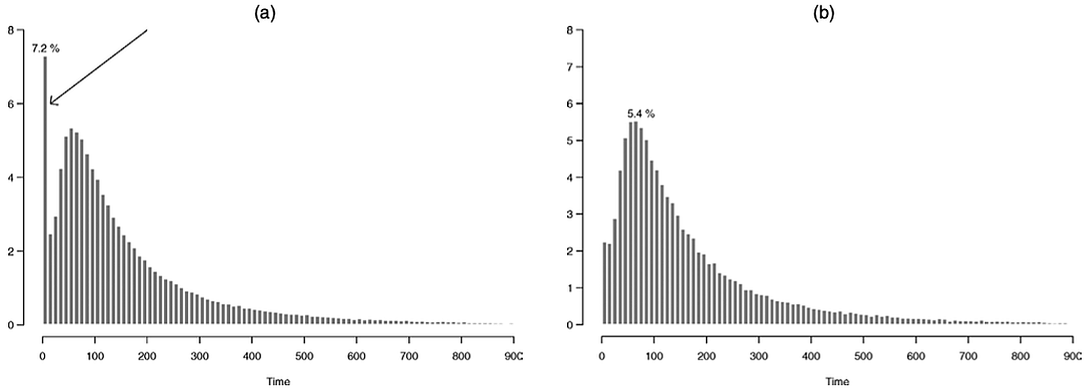
Being able to estimate the parameters in a congestion model is, however, only half the battle. A common mode of failure of stochastic modeling in the real world stems from choosing the wrong model family to begin with. That is, implicit assumptions deep within the constitution of the model can make it so fundamentally deviant from reality from the outset that even sound statistical procedures for estimating the various parameters cannot make up for these more fundamental departures.

In the Erlang examples, there are many such possible assumptions that could, in theory, prevent the model from reliably portraying reality. To name just a few, inter-arrival times may not be exponentially distributed (a fact called out by Erlang himself in the original 1909 paper). The arrival rate is very likely not stationary over time in almost any real-world application. Servers may not work at the same speed. And finally, customers may not patiently wait in line to receive service indefinitely. Some of these assumptions may be necessary evils to cope with in exchange for a relatively elegant and robust mathematical model that is still useful enough, while others may be so egregious as to become deal-breakers. Without further evaluation, a practitioner simply cannot tell one from another.

The study by Brown et al. (2005) offers an interesting case study for evaluating Erlang-style models against real-world operating conditions. The study involves call-by-call operational data from a small bank call center in Israel during the full year of 1999. Within the 1.2 million incoming calls in the dataset, about 750,000 terminate in an automated Voice Responder Unit (VRU) and do not reach any human agents, while the remaining 450,000 requests go past the VRU before queueing for a human agent. Even here, before any analysis begins, we see that the Erlang model may capture only part of a larger real-world system, given that more than 60% of the calls did not even enter this queueing system to begin with. The paper then examines a number of common modeling assumptions against the empirical data. Some of the findings are as one would expect, while others are quite surprising.



(a) Arrival-rate pattern (Figure 1).



(b) Service-time distribution (Figure 2).

Figure 3.1: Empirical diagnostics from (Brown et al., 2005): (a) arrival-rate pattern and (b) service-time distribution.

For starters, two core assumptions in Erlang models are that (a) calls arrive at a constant rate according to a Poisson process, and (b) the time it takes to process each call is exponentially distributed. The data shows that neither assumption matches reality. Figure 3.1a shows the arrival rate over time, and as one would expect, the rate varies greatly over the course of a day. Figure 3.1b shows the service-time distributions over two different months. In both cases, the distribution is not Exponential. For instance, the distribution peaks around 80, whereas an Exponential distribution would have the highest probability density around 0. Upon reflection, this is not too surprising, as one might expect that a phone call to a bank would have some "typical" length that is long enough to resolve an issue. In fact, the paper shows that the empirical service-time distribution is much better approximated by a Lognormal distribution, which tends to have a much heavier tail than an Exponential distribution (i.e., there is a greater chance of very long calls)[1].

Having such a mismatch between the core modeling assumptions and empirical data does not bode well for the validity of Erlang models. For non-stationary arrival rates, the issue is perhaps less severe. Practitioners have long circumvented this problem by simply using the Erlang formula in smaller time windows, typically a 30-minute to 1-hour window. It is believed that during these smaller windows, arrival rates are stable enough to be treated as constant. In other words, to model the congestion level over, say, 12 hours, we would run $12 \times 2 = 24$ separate Erlang formulas. Using the Erlang formula with a non-Exponential service-time distribution is, however, a further stretch, because we lose all the convenience that comes with working with a Markov process, and we now know that congestion dynamics depend on the coefficient of variation of the service-time distribution (the ratio between variance and mean squared), which is very different under the assumed Exponential versus the more realistic Lognormal distribution.[2]

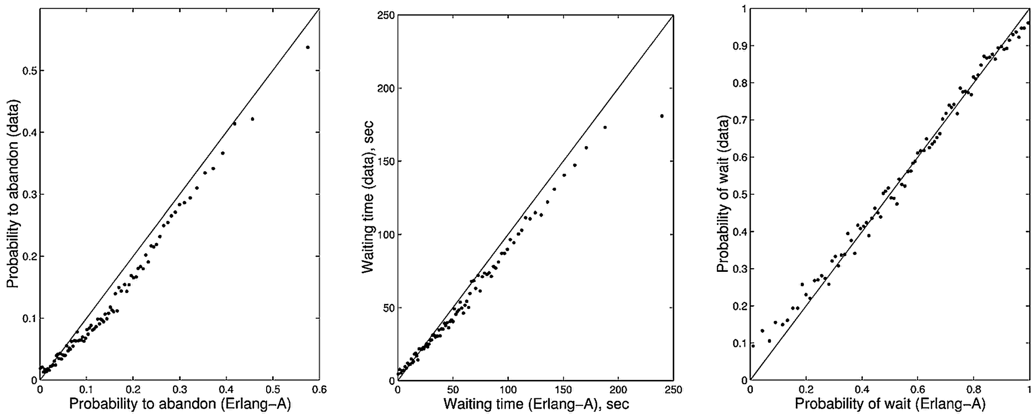


Figure 3.2: Figure 8 in (Brown et al., 2005), comparing modified Erlang-model predictions against observed congestion outcomes.

Given all this mismatch between assumptions and data, one might expect Erlang models to perform poorly at predicting observed congestion metrics. Well... they do not! The researchers then evaluated a modified Erlang model[3] against actual congestion measurements, such as probability of waiting and average waiting times. In Figure 3.2, a total of 3,867 hour-long intervals are partitioned into 45 groups, where each group contains hours with similar congestion loads (arrival rate divided by total service rate), and each dot represents the average measurement from that group. We see that, barring some underestimation of the abandonment probabilities, the Erlang model is generally quite remarkable at producing congestion estimates that closely track empirically observed

[1] For a plausible explanation of lognormality via theories of mathematical psychology, see (Ulrich and Miller, 1993).

[2] See (Whitt, 1993).

[3] They added an additional feature to model customer abandonments while waiting in queues, but other than that the model still closely tracks the original Erlang version.

outcomes. The contrast is striking: despite being built on evidently idealized and "unreal" assumptions, Erlang models still seem to have captured a certain core essence of the system dynamics and are able to deliver accurate predictions.

Wrong Input But Right Output?

Exercise 3.1. Can you think of possible explanations for the apparent paradox in the empirical evaluation of the Erlang model in this section? Specifically, how can the model produce relatively accurate predictions, even though the service-time distributions are Lognormal rather than Exponential, clearly violating some of the core modeling assumptions?

3.3 Evaluating Mechanistic Models and the Presumption of Causality

The discussion above illustrates a general style of model evaluation often used for mechanistic "Erlangian" stochastic models, with Brown et al. (2005) serving as a particularly careful empirical example. By "Erlangian," we mean stochastic models composed of

1. A set of basic structural or distributional assumptions on the underlying random primitives. In this case, think of the random arrival process and the Exponential service-time assumption.

2. A set of clear axioms and rules describing how units and sources of randomness interact with each other. In this case, this corresponds to how users queue in a buffer and how they are served by processors as service completions occur.

To evaluate such a model using empirical data, one would:

1. Compare the empirically observed distribution of the random primitives against the modeling assumptions. In the Erlang example, this amounts to checking whether arrivals follow a Poisson process and whether service times are better characterized as Exponential or Lognormal.

2. Compare the model's final predictions against observed outcomes under similar inputs. The comparison between empirical and predicted congestion levels under different congestion loads, shown in Figure 3.2, provides a concrete example of this type of evaluation.

Mechanistic Connective Fabric

Notice that the above general procedure leaves two key gaps, effectively creating two leaps of faith. First, the empirical analysis does not explicitly verify the axiomatic and logical "connective tissues" that underpin the core operational mechanism of the models. For instance, it is taken as self-evident how a queue works, or that a customer call would first enter a queue, gradually move forward in the queue, and then be transferred into a human-agent session for service. These types of structural assumptions are deeply embedded in the fabric of the stochastic model itself, and are more often argued for or against with qualitative logic than verified with quantitative empirical data.

Why would a designer like Erlang confidently postulate these core connective elements of a model? I suspect one key source of confidence is that the environment in question is designed or engineered by humans, rather than a complex natural phenomenon (think genomics or cancer). Therefore, a practitioner who has accumulated detailed knowledge of the operations of a man-made system, like Erlang, who worked closely with the switching systems at CTC, may be able to attest to the fidelity of the model's mechanism even without direct empirical validation.

Cost of Presumed Causality

A second interesting leap of faith is the question of causality. It is often directly assumed in Erlangian mechanistic models that all interventions applied to the primitives will propagate causally throughout the model. For instance, it is assumed that if we change the number of operators, this will causally impact (in this case improve) the blocking probabilities experienced by average customers. In other words, from the perspective of a practitioner who has full faith in the model, there is little need to conduct further experiments or A/B tests when asking "what-if" questions: the model itself provides a sort of complete sandbox that can answer all what-if questions.

This level of causal knowledge is obviously extremely powerful and is indeed often the appeal of using Erlangian mechanistic models. While they may be reasonable approximations of reality in human-designed systems, in which mechanisms are acutely understood, the detailed mechanisms of more complex human or natural systems are often not possible to capture accurately in syntax. Practitioners who insist on relying only on heavily mechanistic models, built with all the blessings and curses of human insight, may find them producing severely biased results or, in the case of AI, entirely unable to achieve superhuman capabilities.

This is where our story with Erlang takes a pause, and we turn to another emerging area of stochastic-modeling research from around the same period, largely motivated by the natural sciences, that took a somewhat more humble viewpoint. It tends to hold that for questions of macro interest, it is better to set mechanism aside and rely more heavily on experimentation and empiricism to ascertain the impact of actions. In modern lingo, this way of thinking is considered more "end-to-end" or "black box," where capability and accuracy trump understanding of internal mechanisms. We will discuss this part of the development in the next chapter: the age of causal inference.

3.4 Recommended Reading

For primary empirical evidence on whether Erlang-style assumptions hold in a modern service system, see (Brown et al., 2005). For a complementary healthcare example in the same empirical spirit, see (Armony et al., 2015), which studies hospital patient-flow data through a queueing-science lens. The SEELab dataset introduced there has also supported follow-on modeling work, including (Xu and Chan, 2016) and (Li et al., 2023). For original historical context behind the Erlang family itself, see Chapter 2 and (Erlang, 1917).

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